

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Subject:	Cryptology (PEC-CSBS801A)	Semester:	8th Semester
Year:	2024-2025	Exam:	End Semester Examination
UPID:	008440	Paper Code:	CS/B.TECH(N)/EVEN/ SEM-8/8440/2024-2025/1028

Time Allotted: 3 Hours

Full Marks: 70

*The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable.*

GROUP-A (VERY SHORT ANSWER TYPE QUESTION)

Answer any ten of the following:

[1 × 10 = 10]

- (I) What is message authentication? **[1]**
- (II) What are the services provided by PGP services? **[1]**
- (III) Define steganography? **[1]**
- (IV) Define Product cipher? **[1]**
- (V) Which of the following modes of operation in DES is used for operating? **[1]**
- (VI) AES uses a _____ bit block size and a key size of _____ bits. **[1]**
- (VII) What is known as Message authentication code? **[1]**
- (VIII) How many rounds does the AES-192 perform? **[1]**
- (IX) Calculate the GCD of 8376756 and 1921028 using Euclidean algorithm. **[1]**
- (X) For $n = 35$, $e = 5$; $C = 10$, What is the plaintext (using RSA)? **[1]**
- (XI) What is the number of round computation steps in the SHA-256 algorithm? **[1]**
- (XII) Consider the following example: **[1]**
 - Size of Plaintext - 48 bytes
 - Size of MAC - 20 bytes
 - Block Length - 8 bytes
 - How many bytes of padding need to be added to the system?

GROUP-B (SHORT ANSWER TYPE QUESTION)

Answer any three of the following:

[5 × 3 = 15]

2. Explain the differences between MAC and HASH function? [5]
3. What are the differences between passive attack and active attack? [5]
4. Compare Substitution and Transposition techniques. [5]
5. Explain in detail the key generation in AES algorithm? [5]
6. Explain the design criteria of block cipher? [5]

GROUP-C (LONG ANSWER TYPE QUESTION)

Answer any three of the following:

[15 × 3 = 45]

7. (a) What is PGP and S/MIME? Explain them briefly. [8]
(b) Write a short note on IP Sec. [7]
8. (a) State any three requirements for authentication. Define the term message digest. [5]
(b) Define the classes of message authentication function. List the authentication message requirements. [5]
(c) Explain the significance of signature function in Digital Signature Standard (DSS) approach. [5]
9. (a) What is triple encryption? How many keys are used in triple encryption? [5]
(b) Explain the strength of triple DES. [5]
(c) Explain S-DES Structure. [5]
10. (a) Find the secret key shared between user A and user B using Diffie-Hellman algorithm for the following $q=353$; α (primitive root) = 3, $X_A=45$ and $X_B=50$. [7]
(b) Explain Chinese Remainder theorem and find X for the given set of congruent equation using CRT: $X \equiv 2 \pmod{3}$, $X \equiv 3 \pmod{5}$, $X \equiv 2 \pmod{7}$. [8]
11. (a) Write the Euclidean Algorithm. Find $\gcd(1970, 1066)$ using Euclid's algorithm. [5]
(b) Assume that $a = 255$ and $n = 11$. We can find $q = 23$ and $r = 2$ using the division algorithm we have learned in arithmetic. Calculate q and r for $a = -255$ and $n = 11$. [5]
(c) State the five modes of operation of block cipher. [5]

***** END OF PAPER *****